

Summary of Vehicle System

TEST RESULTS

- For smallTRIP QUAD & +GPS



Overview

- ▶ smallTRIP unit was tested and evaluated independently by a specialized third-party company. All technical tests were performed under rigorous circumstances in five main categories:
 - Dust,
 - Mechanical,
 - Temperature/Humidity,
 - Radiation,
 - Electrical-Steady State & Transient Electrical Environment.
- ▶ A part of results from the test series are presented in this document.

Inorganic Dust Test

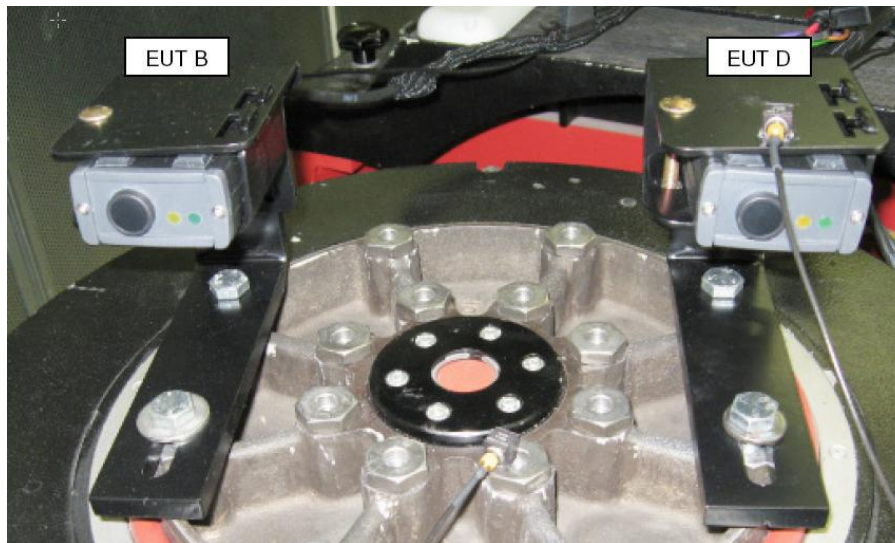
- ▶ Talcum powder for 24h
 - Pass if fully functional after test, dust in enclosure is allowed.



- ◆ Test is passed, the slight leakage can be accepted in the planned use environment.

Vibration Test

- ▶ Sweep Sine and Random Vibration, 3 axis, 8h-axis
 - In both cases the DUT (Device-Under-Test) must be fully functional during and after test.



- ◆ Test is passed for the smallTRIP units, problems with bracket.

Non Operating Mechanical Shock

- ▶ Transit shock and Installation shock.
 - In both cases DUT must be fully functional after test, no visual damage allowed.



- Transit shock

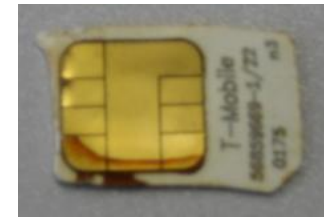
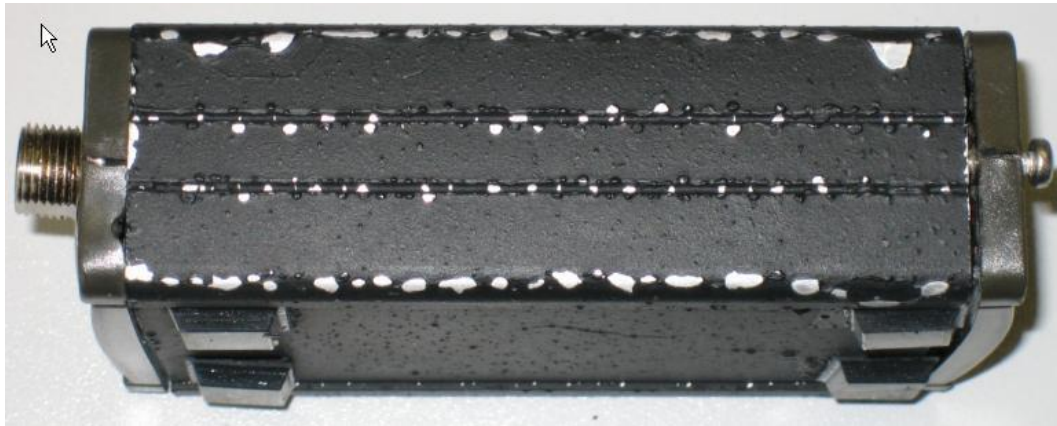


- Installation shock

- ◆ smallTRIP unit / Harness passed both tests.

Combined Environmental Cycle

- ▶ Operating Temperature Humidity, -40°C to 85°C, 24h cycles, 100 cycles.
 - DUT must be fully functional during and after test.



- ◆ smallTRIP unit fails because SIM card is destroyed at high temperatures.
- ◆ When the SIM card is replaced the smallTRIP unit is fully functional again.
- ◆ After 100 days of testing the housing is damaged but still functional.

EMI Test

- ▶ EMI Component Susceptibility
 - ◆ - **some tests showed short Signal loss (fails might have been caused also by GPS receiver)**
 - ◆ - **In general the fail level is low and acceptable.**
- ▶ EMI Emission
 - ◆ **smallTRIP unit passed the test.**

Electrical Transients Environment

- ▶ Jump Forward Voltage (26,5V, 5min, 85°C) - Must be fully functional after test.
 - ◆ **Unit passed the test.**
- ▶ Jump Reverse Voltage (-13,6V, 5min, -40°C) - Must be fully functional during and after test.
 - ◆ **Unit passed the test.**
- ▶ ESD Un-Powered (Handling) (C = 330 pF, R2kOhm, +-15kV) - Must be fully functional after test.
 - ◆ **Unit passed the test.**
- ▶ ESD Powered (C = 150 pF, R2kOhm, +-25kV) - Must be fully functional after test.
 - ◆ **Unit passed the test.**
- ▶ Positive Switching Spikes (100V) - Must be fully functional during and after test.
 - ◆ **Unit passed the test.**